

Solid Waste Management at IIT Gandhinagar

IIT Gandhinagar handles its solid waste on campus: waste is segregated where it is generated, collected daily by a 13-person team, sorted again centrally, and then sent for recycling, turned into biogas or composted.

Overview

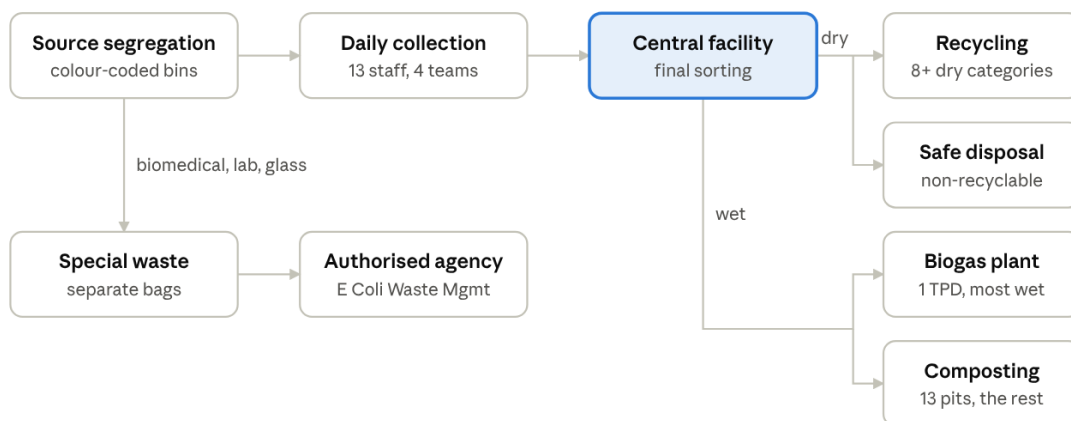
The campus follows a zero-waste approach whose foundation is segregation at the point of generation. Colour-coded bins suited to each location guide users to separate waste correctly, and two further rounds of sorting catch what is missed. Hazardous and specialised waste is kept out of the general stream entirely and handed to an authorised agency, E Coli Waste Management.

The campus generates about 2,000 kg of general waste a day:

Waste type	Generated (kg/day)	Main route
Wet waste	~1,500	Biogas plant (1 TPD), remainder to composting
Dry waste	~500	Recycling (8+ categories), remainder to safe disposal

Waste Flow

Wet waste becomes biogas or compost; dry waste is recycled



General waste passes through collection and central sorting before reaching its end route; special waste bypasses this stream and goes straight to the authorised agency.

Bin System

Buildings get three bins and outdoor areas two, because outdoor waste is smaller in volume and simpler in kind. Every bin is colour-coded and labelled, with informative posters to guide correct disposal.

Bin	Accepts	Buildings (three-bin)	Outdoor and common areas (two-bin)
Green	Organic waste: food waste, kitchen scraps and other biodegradable material	Yes	Yes
Blue	Dry recyclables: paper, plastic, metals	Yes	No
Red	Landfill waste: anything neither recyclable nor compostable (outdoors, all non-organic waste)	Yes	Yes

Three-bin sets are placed in every building: academic blocks, hostels, housing, dining halls and the Research Park. Two-bin sets cover walkways, gardens and open spaces across the 400-acre campus.

Special waste channels. Hazardous and specialised waste is collected in separate colour-coded bags, kept out of the general waste stream, and handled only by an authorised third-party agency, E Coli Waste Management, for compliant treatment and disposal. Laboratory chemicals are not part of this stream.

Bag	Waste
Yellow	Sanitary and biomedical waste
Red	Laboratory consumables (not chemicals)
Blue	Broken glass

Collection

A dedicated SWM staff of 13 is organised each morning into four teams. Two teams stay on their tasks all day; the other two rotate after lunch so a small workforce can cover the whole campus.

Team	Morning	Afternoon
Team 1	Housing area; drops waste at the central hub, then helps Team 2 finish the hostels	Academic area and food outlets, or segregation at the central facility

Team	Morning	Afternoon
Team 2	Hostel area	Academic area and food outlets, or segregation at the central facility (whichever Team 1 does not take)
Team 3 (tempo)	Four dining halls, Research Park, Central Arcade, Sports Complex, Guest House, Security Residence	Same route, all day
Team 4	Biogas plant	Biogas plant, all day

Teams 3 and 4 stay on all day because the tempo route and the biogas plant both need continuous attention. All collected waste is consolidated at the central waste hub.

Worker safety. All SWM staff wear personal protective equipment, including gloves and masks, for every collection and handling task.

Segregation

Waste is segregated in three progressive stages, each catching what the previous one missed.

1. **Primary, at source:** because every building and common area has clearly labelled, colour-coded bins with posters, most residents and users segregate correctly when they dispose of waste.
2. **Secondary, at collection:** collection teams still find some mixed waste, and correct it on the spot wherever possible before it is transported.
3. **Tertiary, at the central facility:** all collected waste is brought to the central waste segregation facility. Dry waste is sorted into more than 8 categories to maximise recycling, and wet waste from every collection point goes directly to the biogas plant.

Processing

Dry waste: recycling. At the central facility, dry waste is sorted into more than 8 categories to maximise recycling. Material that cannot be recycled is disposed of through appropriate channels.

Source reduction: plastic ban at outlets. All eateries, shops and outlets on campus are strictly directed not to use plastic for packaging, food delivery or cutlery. This keeps non-recyclable dry waste from entering the system at all.

Wet waste: biogas. Most of the wet and organic waste is processed at the 1 TPD (tonne per day) biogas plant, which turns food and biodegradable waste into biogas. The biogas fuels a biogas generator that powers the operations of the biogas plant premises.

Wet waste: aerobic composting. The remaining wet waste goes to 13 aerobic composting pits, and the compost is used across the campus in horticulture operations. Compost output is not currently measured. Shredded leaves from horticulture are mixed in to keep the carbon-to-nitrogen ratio around 30:1 .

